Advanced Programming Practice · 21CSC203P

Unit 4 · Session 3 · SLO 2 — Solved Worksheet

Name: AKARSH MURALI AKARSH MURALI | **Reg. No:** RA2511030010067

Unit – 4 : Session – 3 : SLO – 2 | Lambda Expressions and Anonymous Functions

■ I. Programming Task — Custom Higher-Order Function

```
def transform(items, func):
    """Apply func to every element of items and return the new list."""
    return [func(x) for x in items]

# usage
nums = [1, 2, 3, 4]
print(transform(nums, lambda x: x * x)) # [1, 4, 9, 16]
print(transform(nums, lambda x: x + 10)) # [11, 12, 13, 14]
```

■ II. Explanation

- transform() is higher-order because it accepts another function (func) as a parameter.
- The lambda expressions are anonymous functions defined inline — no def or name needed.
- This pattern generalizes map(): any transformation can be plugged in without changing transform() itself.